

Yiling Zha

☎ 437.340.6656 | ✉ tiffany.zha@mail.utoronto.ca | [linkedin.com/in/yilingzha](https://www.linkedin.com/in/yilingzha)

EDUCATION

University of Toronto

Sep. 2022 – Jun. 2027

Bachelor of Applied Science in Electrical and Computer Engineering

Toronto, ON

- CGPA: 3.84
- **Coursework:** Computer Organization, Digital Systems and Electronics, Algorithms Data Structures, Deep Learning
- **Awards:** Dean's Honor List (2022 Fall – 2025 Winter), ECE Honors Standing Award

WORK EXPERIENCE

Synopsys

May. 2025 – Aug. 2026

ASIC Digital Verification Engineer Intern

Markham, ON

- **Owned verification of 5 UALink Accelerator features:** TL packing efficiency, packing limit, backoff, UPLI vdc/wrmsg
- **Developed** testplans and **UVM** block-level verification env with **systemVerilog** and debugged with **Verdi** waveforms.
- **Built** assertion, scoreboard, and stimulus to achieve architectural edge cases and **functional coverage closure**
- **Filed patent** on performance analysis framework with efficiency, throughput, and latency across architecture layers.
- **Gained** PCIe 6.0 PHY verification exposure, with architectural understanding of **PLL, SSC, INL, CTLE, FFE**.

University of Toronto

May. 2024 – Jan. 2026

Research Assistant – Intelligent Sensory Microsystems Lab

Toronto, ON

- **Engineered** digital processing to **generate** high-resolution HDR frames, integrating high-speed and HDR.
- **Supervised** by Prof. Roman Genov and Prof. Amirali Amirsoleimani and **implemented** a Regression-based Flux-to-Digital Conversion readout scheme to **extend dynamic range** beyond conventional ADC architectures.
- **Analyzed** spiking neural network encoding techniques and **mapped** them to **RRAM-based hardware implementations**, studying memory/compute co-location and **array-level constraints**

Shanghai Legend Energy Ltd.

May 2023 – Sep. 2023

Machine Learning Intern

Shanghai, China

- **Developed** LSTM-based time-series models in Python for **load power prediction**, achieving ~90% accuracy, and **analyzed** temporal patterns relevant for **edge/embedded deployment**
- **Designed** feature pipelines for real-time weather data and **trained** Bi-LSTM models for hourly photovoltaic power prediction (~95% accuracy) using BeautifulSoup and Selenium for live data collection.

PROJECTS

FPGA-Based Game System on DE1-SoC (Custom Soft-Core System)

- **Implemented** a two-player fighting game on DE1-SoC using NIOS II Assembly, C, and Verilog.
- **Combined** CPU with **custom peripherals**, including VGA, PS2 keyboard, audio I/O, using **interrupt-driven control**

Travel Planner GIS – C++ Algorithms & Systems

- **Built** a C++ travel planner using OpenStreetMap data and graph algorithms (Dijkstra, A*, Greedy, 3-opt, Simulated Annealing) to **solve** a TSP-like routing problem.
- **Implemented** a multi-threaded backend and interactive GUI (GTK, EZGL, Glade), focusing on **algorithmic performance, memory usage**, and responsiveness for large maps.

SKILLS

Digital Verification: RTL design, Verilog, SystemVerilog, UVM, assertions, coverage-driven verification.

Language and Tools : C++, Verilog, SystemVerilog, Python, Assembly, Java, SQL, Intel Quartus Prime, Verdi